

Anatomy-Aware Masked Image Modeling for Self-Supervised Learning on 3D Brain MRI

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Abstract

Self-supervised pre-training on structural brain MRI can learn transferable representations without diagnostic labels, but standard masked image modeling (MIM) masks patches uniformly at random, ignoring the brain's anatomical organization. Scattered patch reconstruction can be solved from local intensity cues alone, bypassing the inter-regional reasoning most relevant to neuropsychiatric conditions. We propose **anatomy-aware MIM**, which replaces random patch masking with region-level masking derived from cortical parcellations (`aparc+aseg`). By masking entire anatomical regions at a time, the encoder is forced to reconstruct missing brain structures from long-range inter-regional context rather than local neighborhoods, with no additional supervision or parameters. Integrated into a multi-task pre-training framework based on 3D Swin Transformer and evaluated via linear probing, anatomy-aware MIM improves schizophrenia classification over random MIM: balanced accuracy from 64.7% to 66.9%, AUC from 0.767 to 0.771, and recall from 0.398 to 0.451 (+13.3% relative). These results support a broader principle that domain-appropriate masking is a meaningful inductive bias for self-supervised learning in structured biological systems.

Introduction

Large-scale unlabeled neuroimaging datasets offer an opportunity to learn rich brain representations without relying on scarce diagnostic labels. Self-supervised learning (SSL) has emerged as a natural fit for this setting (Tang et al., 2022), and masked image modeling (MIM), where a model learns by reconstructing deliberately hidden input regions, has become one of its most effective instantiations, particularly for transformer architectures (He et al., 2022).

Despite this promise, standard MIM treats the brain like any other image volume by masking patches uniformly at random, disregarding its well-characterized anatomical organization (van Erp et al., 2016). Consequently, reconstruction is often achieved by interpolating from neighboring intensity values, rewarding local texture matching rather than global structural reasoning. For neuroimaging, where clinically relevant signals are distributed across inter-regional morphological relationships, ignoring this structure yields the wrong inductive

bias (Huang et al., 2023) and fails to capture patterns that differentiate clinical populations. What is needed instead is a masking strategy that reflects the brain's own organizational logic.

Structure-aware masking has been explored in adjacent domains. For natural images, (Li et al., 2022) showed that masking semantically coherent object parts, rather than random patches, leads to richer learned representations, improving fine-tuning accuracy on ImageNet by 1.4% over vanilla MAE. In medical imaging, (Tang et al., 2022) proposed domain-tailored proxy tasks for 3D medical image pre-training, demonstrating that anatomy-sensitive objectives yield stronger representations than generic SSL. Yet neither approach translates directly to general-purpose pre-training on 3D structural brain MRI: the former operates on 2D natural images, and the latter targets body-organ CT rather than cortical brain anatomy.

We address this gap with **anatomy-aware MIM**, a simple but principled modification that replaces random patch masking with region-level masking derived from cortical parcellations. Using `aparc+aseg` atlas labels, we mask entire anatomical regions at a time, forcing the model to reconstruct missing brain structures from long-range structural context rather than local neighborhoods. This task-agnostic strategy requires no additional supervision and integrates naturally into existing MIM pipelines. We implement it within the DAMT framework (Kim et al., 2024), a multi-task pre-training approach built on 3D Swin Transformer (Liu et al., 2021), and evaluate downstream performance on schizophrenia classification. Our results support a broader hypothesis: that aligning self-supervised objectives with the intrinsic organization of the data domain is a general principle for learning more meaningful representations in structured biological systems.

Methods

Datasets

Pre-training uses 15,532 T1-weighted brain MRI scans from FOMO300K (Cerri et al., 2025), with all schizophrenia subjects excluded to prevent downstream leakage. For evaluation, we assemble a schizophrenia classification cohort from a non-overlapping FOMO300K subset



and six external datasets: CANDI (Kennedy et al., 2012), CCNMD (Repovš & Barch, 2012), COBRE (Aine et al., 2017), DecNef (Tanaka et al., 2021), MCICShare (Golub et al., 2013), and NUSTAST (Wang et al., 2013), comprising 1,460 healthy controls and 590 patients with schizophrenia ($n = 2,050$). All volumes are processed with FastSurfer (Henschel et al., 2020) to obtain brain-extracted images and `aparc+aseg` parcellations.

Multi-Task Pre-Training Framework

We build on the DAMT framework (Kim et al., 2024), which pre-trains a 3D Swin Transformer (Liu et al., 2021) with encoder depths [2, 2, 18, 2] and embedding dimension 48 on T1-weighted brain MRI volumes of size 128^3 voxels at 1.25 mm isotropic resolution. Pre-training jointly optimizes seven complementary pretext tasks: brain anatomy prediction, brain morphology prediction, radiomics texture prediction, patch location prediction, image rotation prediction, MIM, and contrastive learning. Loss weights across tasks are learned automatically via uncertainty-based weighting (Kendall et al., 2018), avoiding manual tuning and gracefully handling competing gradient signals.

Anatomy-Aware Masked Image Modeling

In standard MIM, 75% of non-overlapping 16^3 -voxel patches are masked uniformly at random, with no regard for the underlying anatomy. Our anatomy-aware variant replaces this with region-level masking derived from the Desikan–Killiany cortical atlas (Desikan et al., 2006) (Figure 1). Each 16^3 patch is assigned the dominant anatomical label of its co-registered parcellations, excluding background. Patches are then grouped by anatomical region, and whole regions are randomly selected and masked until the target ratio ($\approx 75\%$) is reached. This produces spatially contiguous, anatomically coherent masks that force the encoder to reconstruct missing brain structures from long-range inter-regional context, rather than interpolating from immediately neighboring patches. When atlas labels are unavailable, the method falls back to standard random masking, preserving compatibility with the broader pre-training pipeline.

Pre-Training and Evaluation

Both model variants are pre-trained for 100 epochs using AdamW with a learning rate of 5×10^{-4} and cosine annealing. To isolate the effect of masking strategy, we compare two conditions (standard random MIM vs. anatomy-aware MIM) while holding all other pretext tasks and hyperparameters fixed. For downstream evaluation, the pre-trained encoder is frozen and 768-dimensional representations are extracted via adaptive average pooling over the final transformer stage. A linear classifier is then trained on these features using 5-fold cross-validation, following the standard linear probing protocol.

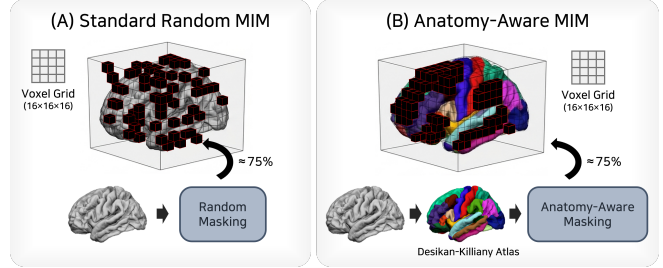


Figure 1: Comparison of (A) standard random MIM and (B) anatomy-aware MIM.

Results

Table 1 reports linear probing performance on schizophrenia classification. Anatomy-aware MIM improves consistently across all metrics: balanced accuracy from 64.7% to 66.9%, AUC from 0.767 to 0.771, and recall from 0.398 to 0.451 (+13.3% relative). The recall gain is especially relevant in clinical screening, where missing true positives carries high cost. Together, these results support the hypothesis that anatomically coherent masking encourages representations encoding the distributed morphological patterns characteristic of schizophrenia.

Table 1: Schizophrenia classification via linear probing: random MIM vs. anatomy-aware MIM.

Metric	Random	Ours
Balanced Accuracy	0.647	0.669
AUC	0.767	0.771
F1-Score	0.482	0.522
Precision	0.609	0.619
Recall	0.398	0.451

Discussion

Anatomy-aware MIM yields consistent gains in schizophrenia classification over random masking, supporting the hypothesis that self-supervised objectives should reflect the intrinsic structure of the data domain. The improvement in recall is particularly meaningful given that schizophrenia is characterized by subtle, distributed morphological alterations spanning multiple cortical and subcortical regions (van Erp et al., 2016), precisely the inter-regional relationships that region-level masking is designed to capture. That these gains arise from a simple modification with no additional parameters or annotations underscores the value of incorporating neuroanatomical priors directly into the pre-training objective.

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